

# Assignment 4 Report

Optimizing Transformer Translation with Ray Tune & Optuna

English → Hindi Neural Machine Translation

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Github Link -

[https://github.com/yadityax/MLOps-Aditya\\_Arun\\_Kumar\\_Yadav-M25CSA001/tree/Assignment-4](https://github.com/yadityax/MLOps-Aditya_Arun_Kumar_Yadav-M25CSA001/tree/Assignment-4)

Hugging Face - <https://huggingface.co/m25csa001/transformer-en-hi/tree/main>

## 1. Baseline Performance (Part 1)

The baseline model was trained end-to-end using the provided *en\_to\_hi.ipynb* notebook without any architectural or hyperparameter modifications. The model trained for **100 epochs** on 13,186 English-Hindi sentence pairs with hardcoded hyperparameters.

| Metric              | Value                                             |
|---------------------|---------------------------------------------------|
| Total Training Time | ~52 minutes                                       |
| Final Training Loss | 0.0974                                            |
| BLEU Score (NLTK)   | 52.47                                             |
| Number of Epochs    | 100                                               |
| Batch Size          | 60                                                |
| Learning Rate       | 1e-4 (Adam)                                       |
| Model Dimensions    | d_model=512, num_heads=8, d_ff=2048, num_layers=6 |
| Dropout             | 0.1                                               |

The baseline loss curve showed steep descent in the first 20 epochs (4.63 → 0.28) followed by slow improvement, indicating the model could converge faster with better hyperparameter choices and a warm-up learning rate schedule.

## 2. Hyperparameter Search (Part 2)

The training loop was refactored into a Ray Tune-compatible **train\_tune(config)** function. Six hyperparameters were selected for tuning using Optuna's TPE (Tree-structured Parzen Estimator) search algorithm, combined with the ASHA scheduler for early pruning of underperforming trials.

| # | Hyperparameter              | Search Range / Choices   | API Used          |
|---|-----------------------------|--------------------------|-------------------|
| 1 | Learning Rate (lr)          | 1e-4 to 5e-4 (log scale) | tune.loguniform() |
| 2 | Batch Size (batch_size)     | {64, 128}                | tune.choice()     |
| 3 | Attention Heads (num_heads) | {4, 8}                   | tune.choice()     |
| 4 | FFN Dimension (d_ff)        | {1024, 2048}             | tune.choice()     |
| 5 | Dropout Rate (dropout)      | 0.1 to 0.3 (uniform)     | tune.uniform()    |

|   |                        |        |               |
|---|------------------------|--------|---------------|
| 6 | Encoder/Decoder Layers | {4, 6} | tune.choice() |
|---|------------------------|--------|---------------|

Ray Tune + Optuna Configuration

**Search Algorithm:** OptunaSearch (metric='loss', mode='min') — uses TPE to intelligently propose hyperparameter combinations based on past trial results.

**Scheduler:** ASHAScheduler (max\_t=40, grace\_period=2, reduction\_factor=2) — terminates unpromising trials after the grace period, keeping only the top 50% at each successive rung. This saved significant compute time compared to running all trials to completion.

**Sweep Settings:** 8 trials, max\_concurrent\_trials=1, GPU-enabled via tune.with\_resources(train\_tune, {'cpu': 2, 'gpu': 1}).

**Additional improvements:** OneCycleLR warm-up scheduler and gradient clipping (clip\_grad\_norm\_=1.0) were added inside train\_tune() to accelerate convergence.

3. Best Configuration Found

After 14.5 minutes of tuning across 8 trials, Optuna identified the following optimal hyperparameter configuration:

| Hyperparameter       | Baseline Value | Best Tuned Value |
|----------------------|----------------|------------------|
| Learning Rate (lr)   | 1e-4           | 2.037e-4         |
| Batch Size           | 60             | 64               |
| Num Attention Heads  | 8              | 8                |
| FFN Dimension (d_ff) | 2048           | 2048             |
| Dropout Rate         | 0.1            | 0.102            |
| Number of Layers     | 6              | 4                |
| d_model              | 512            | 512 (fixed)      |

The most notable finding is that **4 encoder/decoder layers** performed better than 6 within the tuning budget — a shallower model converges faster and avoids overfitting on this relatively small 13K sentence dataset. The slightly higher learning rate (2.037e-4 vs 1e-4) combined with the OneCycleLR scheduler drove faster initial convergence.

4. Final Model Metrics (Part 3 — Efficiency Challenge)

Tuning Sweep Results

| Trial Rank | Loss Epoch Reached | lr | batch_size | num_heads | d_ff | dropout | layers |   |
|------------|--------------------|----|------------|-----------|------|---------|--------|---|
| 1 (Best)   | 0.0804             | 40 | 2.04e-4    | 64        | 8    | 2048    | 0.102  | 4 |
| 2          | 0.1086             | 32 | 4.29e-4    | 128       | 4    | 2048    | 0.129  | 4 |
| 3+         | 5.14+              | 2  | —          | —         | —    | —       | —      | — |

Trials 3–8 were pruned early by ASHA at epoch 2 due to high loss (>5.0), demonstrating the scheduler's effectiveness in saving compute time.

Best Model Retraining (40 Epochs on Best Config)

| Metric          | Baseline (100 epochs) | Best Model (40 epochs) | Improvement      |
|-----------------|-----------------------|------------------------|------------------|
| Training Time   | ~52 minutes           | 4.4 minutes            | ~92% faster      |
| Final Loss      | 0.0974                | 0.0805                 | 17% lower        |
| BLEU Score      | 52.47                 | 73.70                  | +21.23 points    |
| Epochs Required | 100                   | 40                     | 60% fewer epochs |

Loss Curve — Best Model vs Baseline

| Epoch | Best Model Loss | Baseline Loss     |
|-------|-----------------|-------------------|
| 1     | 6.4812          | 4.6352            |
| 5     | 2.9276          | 2.5316            |
| 10    | 1.1871          | 1.09 (approx)     |
| 15    | 0.4531          | 0.55 (approx)     |
| 20    | 0.2325          | 0.2824            |
| 30    | 0.1131          | 0.1793            |
| 40    | 0.0805          | 0.1645 (epoch 34) |

Sample Translations — Best Model

| English Input              | Hindi Output (Predicted)      |
|----------------------------|-------------------------------|
| I love you.                | mujhe tumse pyaar karta hoon. |
| What is your name?         | aapka naam kya hai?           |
| How are you?               | aap kaise ho?                 |
| The weather is nice today. | mausam aaj mausam hai.        |
| She is a good teacher.     | vah ek accha shikshak hai.    |

5. Conclusion

The Ray Tune + Optuna hyperparameter search successfully identified a configuration that **significantly outperforms the baseline** across all metrics. The tuned model achieved a BLEU score of **73.70** compared to the baseline of 52.47 — an improvement of **+21.23 BLEU points** — while using only **40 epochs instead of 100** and completing retraining in just **4.4 minutes vs ~52 minutes**.

The key factors driving this improvement were: (1) Optuna's intelligent search finding a better learning rate (2.04e-4) and reduced model depth (4 layers), (2) the OneCycleLR warm-up scheduler enabling faster convergence, (3) gradient clipping stabilizing training, and (4) the ASHA scheduler efficiently pruning 6 of 8 trials early, reducing total sweep time to just 14.5 minutes.